AN EXHAUSTIVE CENSUS FOR THE $N = 20$ CASE OF THE AOUCHICHE–CAPOROSSI–HANSEN TRICYCLIC-ENERGY CONJECTURE

Abstract. Aouchiche, Caporossi, and Hansen conjectured that every tricyclic graph G on $n = 20$ or $n \geq 22$ vertices satisfies $\mathcal{E}(G) \leq \mathcal{E}(P_n^{6,6,6})$, with equality only for $G \cong P_n^{6,6,6}$. We report an exhaustive binary64 census for the finite case $n = 20$. The census contains 3 288 208 176 isomorphism classes of connected simple graphs with 20 vertices and 22 edges. Its unique computed maximizer is $P_{20}^{6,6,6}$; the computed runner-up is separated from it by approximately 1.3061×10^{-2}. Exact characteristic polynomials for both graphs are given. Because the global comparison was performed in binary64 arithmetic without a uniform interval certificate, the result is stated as an exhaustive numerical verification, not as a proof of the conjecture.

1. The finite case

For a graph G with adjacency eigenvalues $\lambda_1, \ldots, \lambda_n$, its energy is

$$\mathcal{E}(G) = \sum_{i=1}^{n} |\lambda_i|.$$

A connected simple graph is tricyclic when $|E(G)| = |V(G)| + 2$. For $n \geq 20$, let $P_n^{6,6,6}$ be the graph obtained from three copies of C_6 and a path P_{n-18} by adding a single edge between each of two copies of C_6 and one endpoint of the path, and a single edge between the third copy of C_6 and the other endpoint of the path. It has n vertices and $n + 2$ edges; no vertices are identified.

Conjecture 1 (Aouchiche–Caporossi–Hansen [1], in the form restated in [3])**.** *If G is tricyclic on $n = 20$ or $n \geq 22$ vertices, then*

$$\mathcal{E}(G) \leq \mathcal{E}(P_n^{6,6,6}),$$

with equality if and only if $G \cong P_n^{6,6,6}$.

Let $\widehat{\mathcal{E}}(G)$ denote the energy returned by the binary64 eigensolver used in the census.

Computational Result 1. *Among all 3 288 208 176 isomorphism classes of connected simple graphs on 20 vertices and 22 edges, the reported census*

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found $P_{20}^{6,6,6}$ *to be the unique maximizer of* $\widehat{\mathcal{E}}$. *The two largest computed values are approximately*

$$27.2903786151833 \qquad \textit{and} \qquad 27.2773174852287.$$

For an explicit model of $P_{20}^{6,6,6}$, take 6-cycles, in their natural cyclic orders, on $\{0,\dots,5\}$, $\{6,\dots,11\}$, and $\{12,\dots,17\}$, and add

$$\{0,18\}, \qquad \{6,18\}, \qquad \{18,19\}, \qquad \{12,19\}.$$

Its characteristic polynomial is

$$\phi_{P_{20}^{6,6,6}}(x) = (x^2-4)(x^2-1)^4(x^4-7x^2+8)(x^6-7x^4+12x^2-2), \tag{1}$$

and hence

$$\mathcal{E}(P_{20}^{6,6,6}) = 27.2903786151832973536\dots.$$

The runner-up record is isomorphic to the graph

$$Q = P_{20}^{6,6,6} - \{6,18\} + \{5,10\}.$$

It is not isomorphic to $P_{20}^{6,6,6}$; for example, Q has six articulation vertices, whereas $P_{20}^{6,6,6}$ has five. Its characteristic polynomial is

$$\begin{aligned}\phi_Q(x) = (x^2-1)^2(x^{16} &- 20x^{14} + 162x^{12} - 686x^{10} + 1638x^8 \\ &- 2219x^6 + 1621x^4 - 557x^2 + 64),\end{aligned} \tag{2}$$

so that

$$\mathcal{E}(Q) = 27.2773174852287231601\dots$$

and

$$\mathcal{E}(P_{20}^{6,6,6}) - \mathcal{E}(Q) = 0.0130611299545741935\dots.$$

2. Census

The program `geng` from *nauty/Traces* [5] generates one representative of each graph isomorphism class. The census was split into 997 disjoint shards using

`geng -q -c 20 22:22 r/997`, $\qquad 0 \le r < 997.$

The shard counts sum to

$$3\,288\,208\,176,$$

which agrees with the independently tabulated value in OEIS A001436 [6]. A second count-only run with 512 shards gave the same total.

For each graph6 record, the adjacency eigenvalues were computed by Householder tridiagonalization followed by implicit QL iteration, and their absolute values were summed. Besides the two threshold counters displayed next, the scan retained the two largest computed values and one record attaining each; these are the source of the second value in Result 1 and of the runner-up Q. The scan returned

$$\#\{G : \widehat{\mathcal{E}}(G) > \widehat{\mathcal{E}}(P_{20}^{6,6,6}) + 10^{-7}\} = 0,$$
$$\#\{G : |\widehat{\mathcal{E}}(G) - \widehat{\mathcal{E}}(P_{20}^{6,6,6})| \le 10^{-7}\} = 1.$$

The unique record in the second set was verified to be isomorphic to the explicit graph above, and its exact characteristic polynomial agrees with (1). On a comparison sample of 20 000 generated graphs, a second symmetric eigensolver differed in energy by at most 5.15×10^{-13}.

3. Status and scope

The observed gap is approximately 1.31×10^{-2}, far larger than the discrepancies observed in the solver comparison. This is strong numerical evidence for the $n = 20$ case. It is not, however, a uniform error bound over all 3.288 billion comparisons. Result 1 is therefore a statement about the exhaustive floating-point census, not an exact-arithmetic theorem.

A proof-level computer-assisted resolution requires a reproducible archive containing the source, software and build information, per-shard record accounting, and a validated error bound or interval certificate sufficient to separate every competitor from $P_{20}^{6,6,6}$. What is missing is not margin. Every graph in the census satisfies $\|A\|_2^2 \le \sum_i \lambda_i^2 = 2|E| = 44$, hence $\|A\|_2 < 7$, and the standard backward-stability analysis of Householder tridiagonalization followed by QL iteration bounds the error in each computed eigenvalue by $c(n)\,\varepsilon\,\|A\|_2$, with $\varepsilon = 2^{-53}$ and $c(n)$ a modest polynomial in $n = 20$; summed over 20 eigenvalues, any such bound stays many orders of magnitude below the observed gap. The shortfall to a proof therefore lies not in the size of that gap but in two places no margin can cover: the completeness of the generation performed by `geng`, and the applicability of that error model to the code actually executed on each of the 3.288 billion inputs.

The present computation concerns only $n = 20$ and makes no claim for $n \ge 22$. The maximizer itself is not new here: $P_{20}^{6,6,6}$ is the graph already named in the conjecture, so the census confirms a candidate that was on record beforehand rather than discovering one. The conjecture in turn came out of heuristic search, in its case the AutoGraphiX experiments of [1]; earlier searches of the same non-exhaustive kind had been run on related classes, among them the Monte Carlo-type computer experiment of Gutman and Vidović on molecular (n, m)-graphs, which recorded the largest-energy graphs the search found – not provably maximal ones – and offered a tentative guess at the form of the maximal-energy molecular graphs for even n [2]. What the present census adds is exhaustiveness over all 3 288 208 176 isomorphism classes, not the identification of the maximizer. Previous work proves the conjecture inside $G(n; a, b, k)$, the connected bipartite tricyclic graphs on $n \ge 20$ vertices with three vertex-disjoint cycles C_a, C_b, C_k, for most but not all of that class: nine families of exceptions remain in [3], and [4] continues the analysis in the same class. Accordingly, this note records an exhaustive numerical verification of the $n = 20$ case; it does not settle the conjecture.

4. Exact verification

Everything this paper asserts about the two exhibited graphs, together with the census total itself, can be recomputed from the data printed above in exact integer or rational arithmetic, with no floating-point eigensolver and no graph-isomorphism package. That recomputation has been carried out by a second program, written from the definitions and run independently of the one that produced the census. It uses its language's standard library only – no `nauty`, no numerical library – and every decision it takes is made exactly. It reports 64 checks and all 64 pass, but they are not all tests of this paper. A sizeable part of the count tests the program itself: that its subclass enumerator is complete and sound, verified exhaustively against brute force at $n \leq 10$, and that the auxiliary count tables it builds along the way reproduce independently tabulated integer sequences. Those checks license the rest rather than corroborating anything printed here. What the rest do corroborate, and what a referee can accordingly redo from this paper alone, is the following: the explicit model of $P_{20}^{6,6,6}$ and its tricyclicity, bipartiteness, degree sequence and three vertex-disjoint 6-cycles; the articulation counts 5 and 6, and with them the non-isomorphism of $P_{20}^{6,6,6}$ and Q, established twice over, by invariants and by an exhaustive backtracking isomorphism search; the characteristic polynomials (1) and (2), each recomputed from the adjacency matrix and then confirmed a second time by an independent determinant routine; the energies of $P_{20}^{6,6,6}$ and Q and their difference, as rigorous rational enclosures matching every digit printed here; and the census total 3 288 208 176, re-derived by Burnside–Pólya counting with the multiset logarithm, without `geng`. The program and its transcript accompany this note; no claim made here depends on consulting them, since every check just listed can be reproduced from the data printed in this paper.

What this recomputation does not do should be stated plainly, since agreement does not extend to it. It does not repeat the energy scan over all 3 288 208 176 classes, which is thousands of CPU-hours. In place of that it scans exhaustively, with rigorous enclosures, the 8157 isomorphism classes of connected 20-vertex 22-edge graphs all of whose degrees lie in $\{2, 3\}$ – about 2.5×10^{-6} of the census, containing both $P_{20}^{6,6,6}$ and Q, and over which $P_{20}^{6,6,6}$ is again the unique maximizer and Q the runner-up – together with the complete one-edge-swap neighbourhood of $P_{20}^{6,6,6}$ (3363 connected graphs, most of them outside that class), in which no graph other than a copy of $P_{20}^{6,6,6}$ has energy at least $\mathcal{E}(P_{20}^{6,6,6})$. The remaining classes of the census are not rescanned. Nor does it reproduce the shard bookkeeping of the 997- and 512-shard runs, or the two binary64 figures of Section 2 – the counts 0 and 1 at the 10^{-7} threshold and the 5.15×10^{-13} solver agreement – since it replaces floating-point arithmetic by exact enclosures instead of re-measuring it. The conjecture for $n \geq 22$ is outside the scope of both computations. The exact verification agrees with everything it checks; it neither extends the census nor turns it into a proof.

References

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